

Are You Getting What You Pay For?

Auditing Model Substitution in LLM APIs

Software-based detection is fundamentally unreliable — hardware attestation is the provably secure alternative.

arxiv.org/abs/2504.04715

For ML researchers · AI security practitioners · Cloud infrastructure engineers

The Model Substitution Problem

Economic Incentives Drive Covert Swaps

Commercial LLM API providers charge for specific models, but the black-box nature of APIs means users have no guarantee the advertised model is actually being served. Providers face strong economic incentives to covertly substitute cheaper alternatives to reduce GPU costs while maintaining pricing.

1

Smaller Model

e.g., 8B instead of 70B

2

Quantized Variant

FP8/INT8 instead of FP16

3

Fine-Tuned Version

Updated or retrained weights

4

Different Family

Entirely different architecture

Formal Hypothesis Testing Framework

H_0 : Samples come from $P_{\text{spec}}(y|x)$ — the specified model is being served

H_1 : Samples come from $P_{\text{actual}}(y|x)$ — a substitute model is being served

Given samples from the API, can an auditor distinguish the specified model's distribution from an alternative? The answer is far harder than it appears.

FP8 Quantization Preserves Most Accuracy

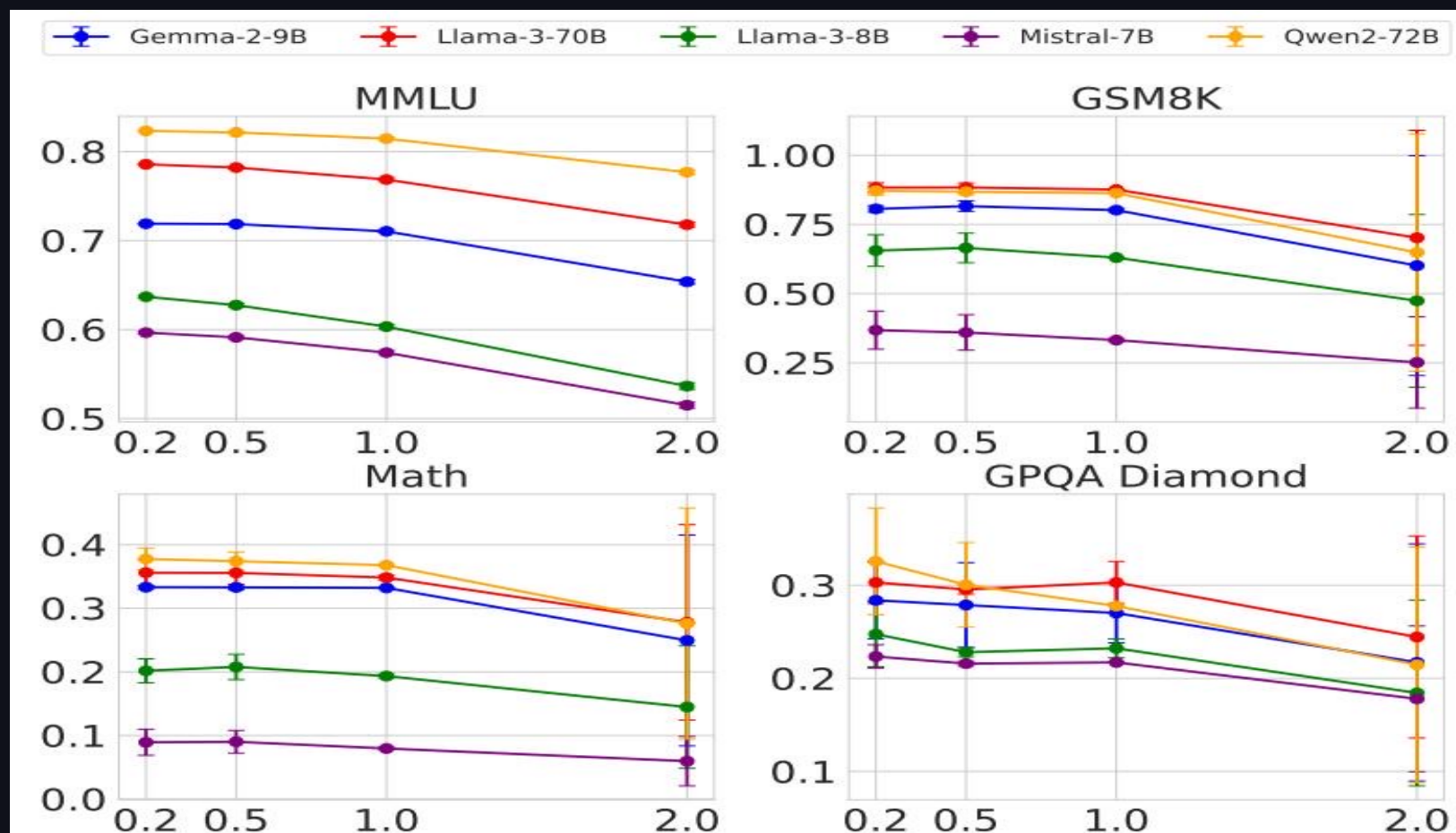
Making Substitution Hard to Detect

Model	MMLU	GSM8K	MATH	GPQA Diamond
Llama-3-8B-Instruct	62.69 ± 0.18	61.14 ± 3.47	20.65 ± 3.43	22.62 ± 0.22
Llama-3-8B-Instruct-FP8	62.43 ± 0.26	60.90 ± 4.10	14.91 ± 2.49	20.14 ± 0.24
Llama-3-70B-Instruct	78.05 ± 0.08	88.06 ± 1.44	35.69 ± 1.33	29.60 ± 0.51
Llama-3-70B-Instruct-FP8	77.88 ± 0.13	87.35 ± 1.24	35.75 ± 1.16	33.12 ± 0.30
Gemma-2-9b-it	71.86 ± 0.08	81.80 ± 1.35	33.41 ± 0.28	28.64 ± 2.97
Gemma-2-9b-it-FP8	71.92 ± 0.11	79.41 ± 1.14	32.53 ± 0.34	27.81 ± 3.22
Qwen2-72B-Instruct	82.18 ± 0.08	86.72 ± 1.00	37.39 ± 1.41	29.93 ± 2.71
Qwen2-72B-Instruct-FP8	81.98 ± 0.08	86.82 ± 0.97	37.67 ± 1.39	31.08 ± 1.96
Mistral-7B-Instruct-v0.3	59.15 ± 0.10	35.90 ± 4.54	8.94 ± 1.24	21.60 ± 0.17
Mistral-7B-Instruct-v0.3-FP8	58.77 ± 0.13	32.20 ± 4.02	7.68 ± 1.23	22.72 ± 0.19

Key insight: FP8 scores fall within the error bars of the original on MMLU and GSM8K — detection via benchmarks alone is unreliable.

Higher Temperature Amplifies Gaps but Increases Noise

Temperature Scaling Degrades Detection Signal



Observations

At $T = 0.2 - 0.5$, accuracy differences between original and FP8 models are minimal.

At $T = 2.0$, all models degrade significantly, and the gap between original and substitute narrows or becomes noisy. Higher temperature adds sampling randomness that obscures the already-subtle substitution signal.

Implication

Auditors cannot reliably pick a temperature that both amplifies the signal and maintains statistical power.

Log Probability Fingerprinting Fails

Inference Nondeterminism Defeats Metadata-Based Detection

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49.80
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50.20

Root Cause

The same Llama-3-8B model produces different log probabilities depending on:

- Software: transformers 4.40 vs 4.50, vLLM 0.6.2 vs 0.8.2
- Hardware: A100 vs H100 GPUs

Fatal Implication

An auditor cannot distinguish legitimate infrastructure variation from actual model substitution.

Log-prob fingerprinting is fundamentally unreliable in production settings.

Randomized Routing at Low Rates

Makes Statistical Detection Query-Prohibitive

Adversarial Strategy: The provider routes only a fraction p of queries to the cheaper substitute model, mixing substitute responses with genuine ones.

Mixed Distribution

$$P_{\text{mixed}}(y|x) = (1 - p) \cdot P_{\text{spec}}(y|x) + p \cdot P_{\text{sub}}(y|x)$$

As $p \rightarrow 0$, $P_{\text{mixed}} \rightarrow P_{\text{spec}}$

Adversarial Advantage

The provider can further combine randomized routing with benchmark evasion — detecting known audit prompts and routing them to the genuine model while serving substitutes for regular traffic. This makes software-only detection fundamentally query-prohibitive.

Detection Cost Escalation



Software-Only Auditing Methods Are Fundamentally Unreliable

Method Type	Weakness	Practical Limitation
Text-Based Statistical Tests (e.g., MMD)	Require impractically large query budgets; fail against subtle substitutions like FP8	Cannot distinguish FP8 from original when scores fall within error bars
Log-Probability Fingerprinting	Defeated by inference nondeterminism across software & hardware	Cannot differentiate legitimate infrastructure variation from actual model substitution
Adversarial Countermeasures	Randomized routing & benchmark evasion undermine all passive detection	As $p \rightarrow 0$, detection requires exponentially more queries; benchmark gaming goes undetected

Verdict: No software-only method can reliably detect model substitution under realistic adversarial conditions.

Trusted Execution Environments

Provable Cryptographic Model Integrity

TEEs are hardware-secured enclaves that provide cryptographic attestation of the code and data being executed. The provider can prove that specific model weights are loaded and the correct inference code is running.

Provable

Cryptographic attestation guarantees model integrity — not a statistical test, but a mathematical proof.

Efficient

Only modest performance overhead compared to ZKPs, which are computationally prohibitive for large models.

Robust

Resistant to all software-based evasion strategies: randomized routing, benchmark evasion, and framework variation.

Software vs. Hardware Attestation

Statistical Tests

Probabilistic, gameable



Log-Prob Fingerprinting

Defeated by nondeterminism



Zero-Knowledge Proofs

Computationally prohibitive



TEEs (NVIDIA H100)

Provable + practical + deployable



NVIDIA confidential computing on H100 GPUs provides a practical deployment path today.

Key Takeaways

Close the Verification Gap with Hardware Attestation

1 Model substitution is economically motivated and practically undetectable via software

Providers save significant GPU costs by swapping FP16 models with FP8 variants or smaller architectures — and current software methods cannot reliably catch this.

2 FP8 quantization barely affects benchmark scores

Performance drops fall within error bars on MMLU and GSM8K, making benchmark-based auditing unreliable for detecting the most cost-effective substitution.

3 Inference nondeterminism kills log-probability fingerprinting

Different software frameworks (transformers, vLLM) and GPU hardware (A100, H100) produce varying log probabilities, making metadata-based detection impossible.

4 Randomized substitution defeats statistical tests

As the mixing probability $p \rightarrow 0$, the mixed distribution converges to the specified model, requiring exponentially more queries — rendering statistical detection impractical.

5 TEEs are the only currently viable path to verifiable LLM APIs

Hardware attestation provides provable, efficient, and robust model integrity guarantees — representing the most actionable solution available today.

Call to Action: The LLM API ecosystem must adopt hardware-based attestation to ensure verifiable inference. The trust gap cannot be closed by software alone.